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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 27th July Morning Shift

JEE Main Question Paper with Answer Key

27 Jul 2021	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	25 questions
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Out of syllabus	18 questions
Answer key	at the end

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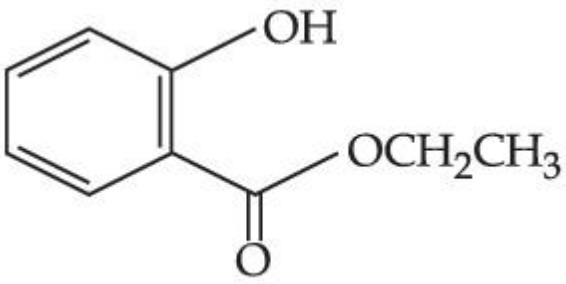
Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

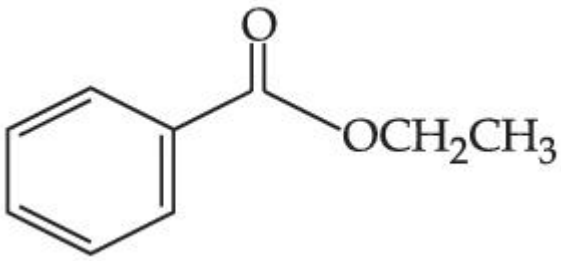
Open solution

Which one of the following compounds will give orange precipitate when treated with 2, 4-dinitrophenyl hydrazine?

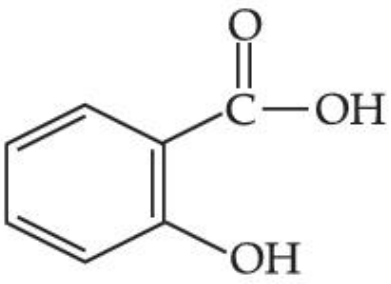
(A)



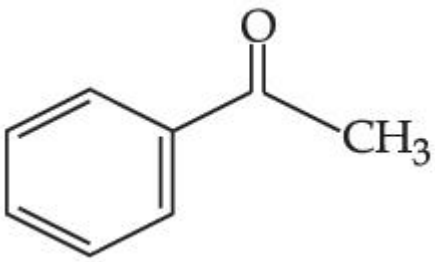
(B)



(C)



(D)



Q2. MCQ (Single Correct Answer)

Open solution

The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$, respectively, are :

(A) 7, 5 and 6 (B) 5, 4 and 3 (C) 5, 3 and 4 (D) 6, 4 and 5

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Q3. MCQ (Single Correct Answer)

For a reaction of order n, the unit of the rate constant is :

(A) $mol^{1-n}\,L^{1-n}\,s$ (B) $mol^{1-n}\,L^{2n}\,s^{-1}$ (C) $mol^{1-n}\,L^{n-1}\,s^{-1}$ (D) $mol^{1-n}\,L^{1-n}\,s^{-1}$

Open solution

Q4. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : Aniline is less basic than acetamide.

Statement II : In aniline, the lone pair of electrons on nitrogen atom is delocalized over benzene ring due to resonance and hence less available to a proton.

Choose the most appropriate option;

(A) Statement I is true but statement II is false.

(B) Statement I is false but statement II is true.

(C) Both statement I and statement II are true.

(D) Both statement I and statement II are false.

Open solution

Q5. MCQ (Single Correct Answer)

The type of hybridisation and magnetic property of the complex $[MnCl_6]^{3-}$, respectively, are :

(A) sp^3d^2 and diamagnetic (B) d^2sp^3 and diamagnetic

(C) d^2sp^3 and paramagnetic (D) sp^3d^2 and paramagnetic

Open solution

Q6. MCQ (Single Correct Answer)

The number of geometrical isomers found in the metal complexes $[PtCl_2(NH_3)_2]$, $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are :

(A) 1, 1, 1, 1 (B) 2, 1, 2, 2 (C) 2, 0, 2, 2 (D) 2, 1, 2, 1

Open solution

Q7. MCQ (Single Correct Answer)

Presence of which reagent will affect the reversibility of the following reaction, and change it to an irreversible reaction :

$$CH_4 + I_2 \underset{\text{Reversible}}{\overset{h\nu}{\rightleftharpoons}} CH_3 - I + HI$$

(A) HOCl (B) dilute HNO_2

(C) Liquid NH_3 (D) Concentrated HIO_3

Open solution

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Q8. MCQ (Single Correct Answer)

Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide?

Open solution

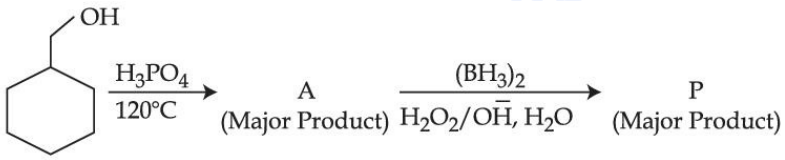
(A) Seliwanoff's test

(B) Iodine test

(C) Barfoed test

(D) Tollens' test

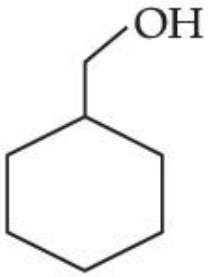
Q9. MCQ (Single Correct Answer)



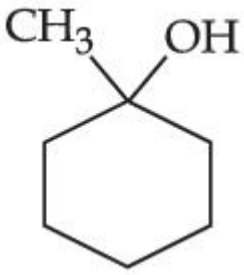
Open solution

Consider the above reaction and identify the Product P :

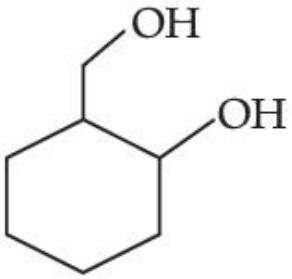
(A)



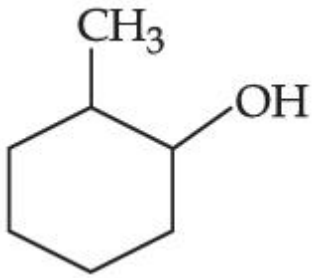
(B)



(C)

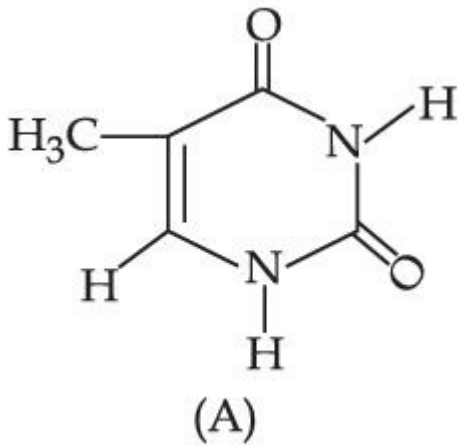


(D)



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Q10. MCQ (Single Correct Answer)



(A)

Open solution

The compound 'A' is a complementary base of _____ in DNA strands.

(A) Uracil **(B)** Guanine **(C)** Adenine **(D)** Cytosine

Q11. MCQ (Single Correct Answer)

Staggered and eclipsed conformers of ethane are :

(A) Polymers **(B)** Rotamers **(C)** Enantiomers **(D)** Mirror images

Open solution

Q12. MCQ (Single Correct Answer)

Match List - I with List - II :

Open solution

	List - I		List - II
(a)	NaOH	(i)	Acidic
(b)	$Be(OH)_2$	(ii)	Basic
(c)	$Ca(OH)_2$	(iii)	Amphoteric
(d)	$B(OH)_3$		
(e)	$Al(OH)_3$		

Choose the most appropriate answer from the options given below

(A) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii)

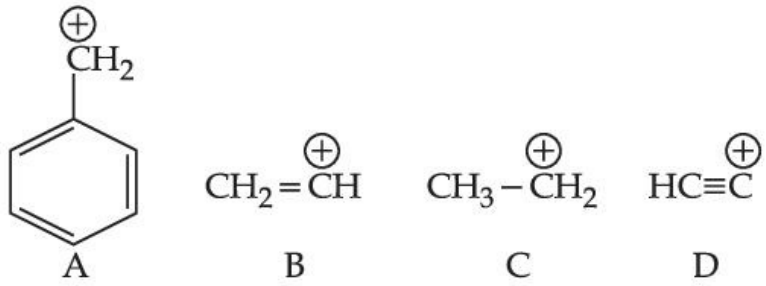
(B) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)

(C) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii)

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Q13. MCQ (Single Correct Answer)



A B C D

The correct order of stability of given carbocation is :

(A) A > C > B > D **(B)** D > B > C > A **(C)** D > B > A > C **(D)** C > A > D > B

Open solution

Q14. Numerical

The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is _____ m. (Round off to the Nearest Integer)

[Use : Atomic mass : Na : 23.0 u, O : 16.0 u, H : 1.0 u, Density of H_2O : 1.0 g cm^{-3}]

Open solution

Q15. Numerical

The conductivity of a weak acid HA of concentration 0.001 mol L^{-1} is 2.0×10^{-5} S cm^{-1}. If Λ_m^o(HA) = 190 S cm^2 mol^{-1}, the ionization constant (K_a) of HA is equal to _____ $\times 10^{-6}$. (Round off to the Nearest Integer)

Open solution

Q16. Numerical

1.46 g of a biopolymer dissolved in a 100 mL water at 300 K exerted an osmotic pressure of 2.42×10^{-3} bar.

The molar mass of the biopolymer is _____ $\times 10^4$ g mol^{-1}. (Round off to the Nearest Integer)

[Use : R = 0.083 L bar mol^{-1} K^{-1}]

Open solution

Q17. Numerical

An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with $AgNO_3$ [Carius Method], the percentage of chlorine in compound A is _____ when it forms 0.3849 g of AgCl. (Round off to the Nearest Integer)

(Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)

Open solution

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Q18. Numerical

The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____.

Open solution

Q19. Numerical

In gaseous triethyl amine the "−C−N−C−" bond angle is _____ degree.

Open solution

Q20. Numerical

For water at 100° C and 1 bar,

$\Delta_{vap} H - \Delta_{vap} U =$ _____ $\times 10^2$ J mol^{-1}. (Round off to the Nearest Integer)

[Use : R = 8.31 J mol^{-1} K^{-1}]

[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ treated as an ideal gas]

Open solution

Q21. Numerical

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$K_c = 1.844$

3.0 moles of PCl_5 is introduced in a 1 L closed reaction vessel at 380 K. The number of moles of PCl_5 at equilibrium is _____ $\times 10^{-3}$. (Round off to the Nearest Integer)

Open solution

Q22. Numerical

The difference between bond orders of CO and $NO^{\oplus}$ is $\frac{x}{2}$ where x = _____. (Round off to the Nearest Integer)

Open solution

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Mathematics 25 questions

Q23. MCQ (Single Correct Answer)

If the mean and variance of the following data : 6, 10, 7, 13, a, 12, b, 12 are 9 and $\frac{37}{4}$

respectively, then (a $-$ b)^2 is equal to :

(A) 24 (B) 12 (C) 32 (D) 16

Open solution

Q24. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\vec{a}+\vec{b}\right) \times \left(\left(\vec{a} \times \left(\left(\vec{a}-\vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ is equal to :

(A) $5(34\hat{i} - 5\hat{j} + 3\hat{k})$ (B) $7(34\hat{i} - 5\hat{j} + 3\hat{k})$

(C) $7(30\hat{i} - 5\hat{j} + 7\hat{k})$ (D) $5(30\hat{i} - 5\hat{j} + 7\hat{k})$

Open solution

Q25. MCQ (Single Correct Answer)

The value of the definite integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{(1+e^{x\cos x})(\sin^4 x+\cos^4 x)}$ is equal to :

(A) $-\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$ (C) $-\frac{\pi}{4}$ (D) $\frac{\pi}{\sqrt{2}}$

Open solution

Q26. MCQ (Single Correct Answer)

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z - 3 - 2i|^2 = 8\}$

$S_2 = \{z \in C : \text{Re}(z) \geq 5\}$ and

$S_3 = \{z \in C : |z - \bar{z}| \geq 8\}$.

Then the number of elements in $S_1 \cap S_2 \cap S_3$ is equal to :

(A) 1 (B) 0 (C) 2 (D) Infinite

Open solution

Q27. MCQ (Single Correct Answer)

If the area of the bounded region
$R = \{(x, y) : \max\{0, \log_e x\} \leq y \leq 2^x, \frac{1}{2} \leq x \leq 2\}$ is ,
$\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, then the value of $(\alpha + \beta - 2\lambda)^2$ is equal to :

(A) 8 (B) 2 (C) 4 (D) 1

Open solution

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Q28. MCQ (Single Correct Answer)

A ray of light through (2, 1) is reflected at a point P on the y-axis and then passes through the point (5, 3). If this reflected ray is the directrix of an ellipse with eccentricity $\frac{1}{3}$ and the distance of the nearer focus from this directrix is $\frac{8}{\sqrt{53}}$, then the equation of the other directrix can be :

Open solution

(A) 11x + 7y + 8 = 0 or 11x + 7y − 15 = 0

(B) 11x − 7y − 8 = 0 or 11x + 7y + 15 = 0

(C) 2x − 7y + 29 = 0 or 2x − 7y − 7 = 0

(D) 2x − 7y − 39 = 0 or 2x − 7y − 7 = 0

Q29. MCQ (Single Correct Answer)

If the coefficients of x^7 in $\left(x^2+\frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x-\frac{1}{bx^2}\right)^{11}$, $b \neq 0$, are equal, then the value of b is equal to :

Open solution

(A) 2 (B) -1 (C) 1 (D) -2

Q30. MCQ (Single Correct Answer)

If $\sin\theta+\cos\theta=\frac{1}{2}$, then 16(sin(2$\theta$) + cos(4$\theta$) + sin(6$\theta$)) is equal to :

Open solution

(A) 23 (B) -27 (C) -23 (D) 27

Q31. MCQ (Single Correct Answer)

Let $A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$. If $A^{-1} = \alpha I + \beta A$, $\alpha, \beta \in R$, I is a 2×2 identity matrix then $4(\alpha-\beta)$ is equal to :

Open solution

(A) 5 (B) $\frac{8}{3}$ (C) 2 (D) 4

Q32. MCQ (Single Correct Answer)

Let $f:\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\to R$ be defined as $f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}} & , \quad -\frac{\pi}{4}<x<0\\ b & , \quad x=0\\ e^{\cot 4x/\cot 2x} & , \quad 0<x<\frac{\pi}{4}\end{cases}$

Open solution

If f is continuous at x = 0, then the value of 6a + b^2 is equal to :

(A) 1 − e (B) e − 1 (C) 1 + e (D) e

Q33. MCQ (Single Correct Answer)

Let y = y(x) be solution of the differential equation

$\log_e\left(\frac{dy}{dx}\right)=3x+4y$, with y(0) = 0.

Open solution

If $y\left(-\frac{2}{3}\log_e 2\right)=\alpha\log_e 2$, then the value of α is equal to :

(A) $-\frac{1}{4}$ (B) $\frac{1}{4}$ (C) 2 (D) $-\frac{1}{2}$

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Q34. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be a function such that f(2) = 4 and f'(2) = 1. Then, the value of $\lim_{x \to 2} \frac{x^2 f(2) - 4f(x)}{x-2}$ is equal to :

(A) 4 (B) 8 (C) 16 (D) 12

Open solution

Q35. MCQ (Single Correct Answer)

Let P and Q be two distinct points on a circle which has center at C(2, 3) and which passes through origin O. If OC is perpendicular to both the line segments CP and CQ, then the set {P, Q} is equal to :

(A) {(4, 0), (0, 6)} (B) $\{(2+2\sqrt{2}, 3-\sqrt{5}), (2-2\sqrt{2}, 3+\sqrt{5})\}$

(C) $\{(2+2\sqrt{2}, 3+\sqrt{5}), (2-2\sqrt{2}, 3-\sqrt{5})\}$ (D) {(−1, 5), (5, 1)}

Open solution

Q36. MCQ (Single Correct Answer)

Let α, β be two roots of the

equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to

(A) 10 (B) 100 (C) 50 (D) 160

Open solution

Q37. MCQ (Single Correct Answer)

The probability that a randomly selected 2-digit number belongs to the set {n $\in$ N : $(2^n - 2)$ is a multiple of 3} is equal to :

(A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Open solution

Q38. MCQ (Single Correct Answer)

Let $A = \{(x, y) \in R \times R | 2x^2 + 2y^2 - 2x - 2y = 1\}$, $B = \{(x, y) \in R \times R | 4x^2 + 4y^2 - 16y + 7 = 0\}$ and $C = \{(x, y) \in R \times R | x^2 + y^2 - 4x - 2y + 5 \leq r^2\}$.

Then the minimum value of |r| such that $A \cup B \subseteq C$ is equal to

(A) $\frac{3+\sqrt{10}}{2}$ (B) $\frac{2+\sqrt{10}}{2}$ (C) $\frac{3+2\sqrt{5}}{2}$ (D) $1 + \sqrt{5}$

Open solution

Q39. Numerical

For real numbers α and β, consider the following system of linear equations :

x + y − z = 2, x + 2y + αz = 1, 2x − y + z = β. If the system has infinite solutions, then $\alpha + \beta$ is equal to _____.

Open solution

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Q40. Numerical

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b}$ and $\vec{c} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l, then the value of $3l^2$ is equal to _____.

Open solution

Q41. Numerical

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to _____.

Open solution

Q42. Numerical

Let the domain of the function

$f(x) = \log_4\left(\log_5\left(\log_3(18x - x^2 - 77)\right)\right)$ be (a, b). Then the value of the integral $\int_a^b \frac{\sin^3 x}{\sin^3 x + \sin^3(a+b-x)} dx$ is equal to _____.

Open solution

Q43. Numerical

Let $f(x) = \begin{vmatrix} \sin^2 x & -2 + \cos^2 x & \cos 2x \\ 2 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1 + \cos 2x \end{vmatrix}, x \in [0, \pi]$. Then the maximum value of f(x) is equal to _____.

Open solution

Q44. Numerical

Let $F : [3, 5] \to R$ be a twice differentiable function on (3, 5) such that

$F(x) = e^{-x} \int_3^x (3t^2 + 2t + 4F'(t))dt$. If $F'(4) = \frac{\alpha e^\beta - 224}{(e^\beta - 4)^2}$, then $\alpha + \beta$ is equal to _____.

Open solution

Q45. Numerical

Let S = {1, 2, 3, 4, 5, 6, 7}. Then the number of possible functions f : S $\to$ S
such that f(m . n) = f(m) . f(n) for every m, n $\in$ S and m . n $\in$ S is equal to _____.

Open solution

Q46. Numerical

If $y = y(x), y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation
$\sec y \frac{dy}{dx} - \sin(x + y) - \sin(x - y) = 0$, with y(0) = 0, then $5y'\left(\frac{\pi}{2}\right)$ is equal to _____.

Open solution

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Q47. Numerical

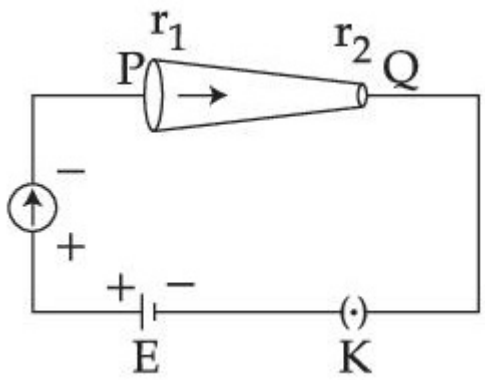
Let $f : [0, 3] \to R$ be defined by $f(x) = \min\{x - [x], 1 + [x] - x\}$ where [x] is the greatest integer less than or equal to x. Let P denote the set containing all x $\in$ [0, 3] where f i discontinuous, and Q denote the set containing all x $\in$ (0, 3) where f is not differentiable. Then the sum of number of elements in P and Q is equal to _____.

Open solution

Physics 25 questions

Q48. MCQ (Single Correct Answer)

In the given figure, a battery of emf E is connected across a conductor PQ of length 'l' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).



Open solution

Choose the correct option as one moves from P to Q :

(A) Drift velocity of electron increases.
(B) Electric field decreases.
(C) Electron current decreases.
(D) All of these

Q49. MCQ (Single Correct Answer)

The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy 2×10^{-9} J per molecules is :

(A) 0.75×10^{11} (B) 3×10^{11} (C) 1.5×10^{11} (D) 6×10^{11}

Open solution

Q50. MCQ (Single Correct Answer)

The relative permittivity of distilled water is 81. The velocity of light in it will be :

(Given μ_r = 1)

(A) 4.33×10^7 m/s (B) 2.33×10^7 m/s (C) 3.33×10^7 m/s (D) 5.33×10^7 m/s

Open solution

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Q51. MCQ (Single Correct Answer)

Open solution

List-I	List-II
(a) MI of the rod (length L, Mass M, about an axis $\perp$ to the rod passing through the midpoint)	(i) $8ML^2/3$
(b) MI of the rod (length L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(ii) $ML^2/3$
(c) MI of the rod (length 2L, Mass M, about an axis $\perp$ to the rod passing through its midpoint)	(iii) $ML^2/12$
(d) MI of the rod (Length 2L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(iv) $2ML^2/3$

Choose the correct answer from the options given below:

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

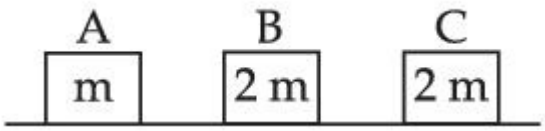
(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q52. MCQ (Single Correct Answer)

Open solution

Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2m and 2m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is :



(A) 6 m/s

(B) 9 m/s

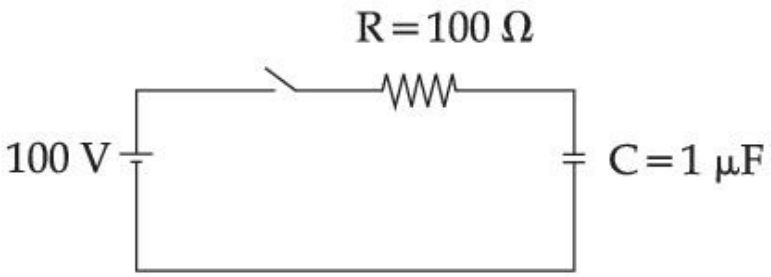
(C) 4 m/s

(D) 3 m/s

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Q53. MCQ (Single Correct Answer)

Open solution



A capacitor of capacitance C = 1 μF is suddenly connected to a battery of 100 volt through a resistance R = 100 Ω. The time taken for the capacitor to be charged to get 50 V is :

[Take ln 2 = 0.69]

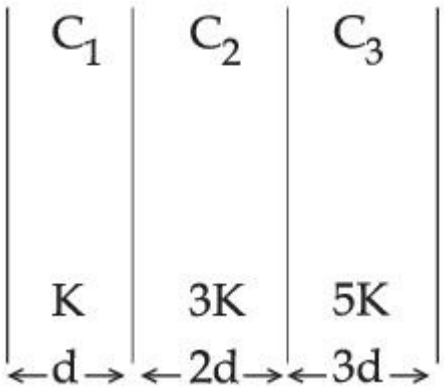
(A) 1.44×10^{-4} s (B) 3.33×10^{-4} s (C) 0.69×10^{-4} s (D) 0.30×10^{-4} s

Q54. MCQ (Single Correct Answer)

Open solution

In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be :

(Given area of plate = A)



(A) $\frac{15}{34}\frac{K\varepsilon_0 A}{d}$ (B) $\frac{15}{6}\frac{K\varepsilon_0 A}{d}$

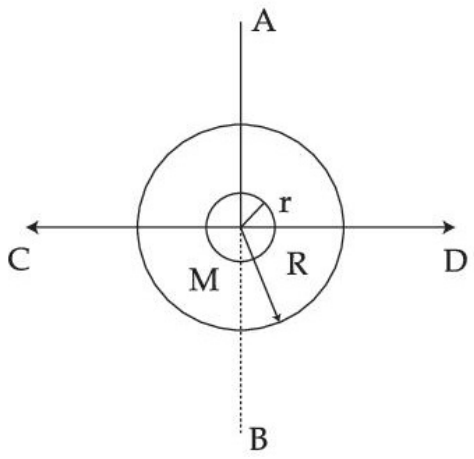
(C) $\frac{25}{6}\frac{K\varepsilon_0 A}{d}$ (D) $\frac{9}{6}\frac{K\varepsilon_0 A}{d}$

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Q55. MCQ (Single Correct Answer)

Open solution

The figure shows two solid discs with radius R and r respectively. If mass per unit area is same for both, what is the ratio of MI of bigger disc around axis AB (Which is $\perp$ to the plane of the disc and passing through its centre) to MI of smaller disc around one of its diameters lying on its plane? Given 'M' is the mass of the larger disc. (MI stands for moment of inertia)



(A) $R^2 : r^2$ (B) $2r^4 : R^4$ (C) $2R^2 : r^2$ (D) $2R^4 : r^4$

Q56. MCQ (Single Correct Answer)

Open solution

In Young's double slit experiment, if the source of light changes from orange to blue then :

(A) the central bright fringe will become a dark fringe.

(B) the distance between consecutive fringes will decrease.

(C) the distance between consecutive fringes will increases.

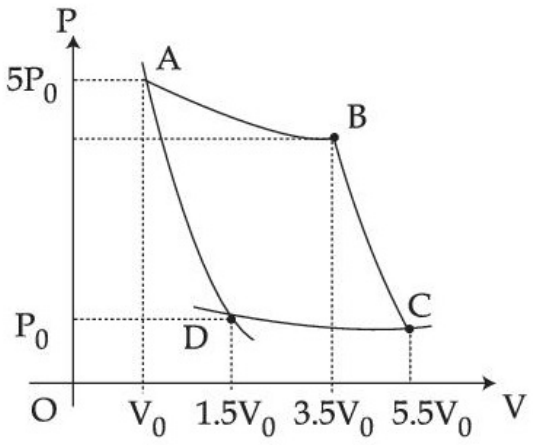
(D) the intensity of the minima will increase.

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Q57. MCQ (Single Correct Answer)

Open solution

In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the process A $\rightarrow$ B and C $\rightarrow$ D are T_1 and T_2 ($T_1 > T_2$) respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

(A) $W_{AB} = W_{DC}$ (B) $W_{AD} = W_{BC}$ (C) $W_{BC} + W_{DA} > 0$ (D) $W_{AB} < W_{CD}$

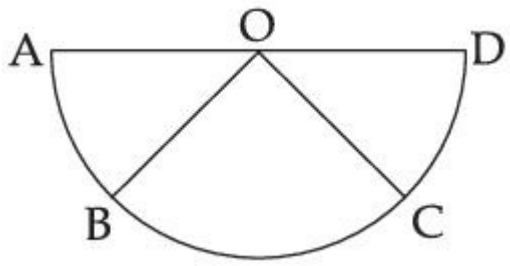
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Q58. MCQ (Single Correct Answer)

Open solution

Assertion A : If A, B, C, D are four points on a semi-circular arc with centre at 'O' such that $\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{CD}\right|$, then $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$

Reason R : Polygon law of vector addition yields $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AD} = 2\overrightarrow{AO}$



In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct.

(B) A is not correct but R is correct.

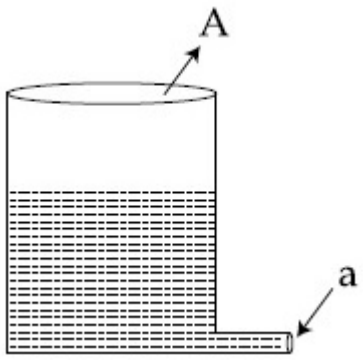
(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is not the correct explanation of A.

Q59. MCQ (Single Correct Answer)

Open solution

A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is (a < < A) :



(A) $\frac{A}{2a}$ (B) None of these (C) $\frac{2a}{A}$ (D) $\frac{a}{A}$

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Q60. MCQ (Single Correct Answer)

A particle starts executing simple harmonic motion (SHM) of amplitude 'a' and total energy E. At any instant, its kinetic energy is $\frac{3E}{4}$ then its displacement 'y' is given by :

(A) y = a (B) $y = \frac{a}{\sqrt{2}}$ (C) $y = \frac{a\sqrt{3}}{2}$ (D) $y = \frac{a}{2}$

Open solution

Q61. MCQ (Single Correct Answer)

Two capacitors of capacities 2C and C are joined in parallel and charged up to potential V. The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K. The potential difference across the capacitors will now be :

(A) $\frac{V}{K+2}$ (B) $\frac{V}{K}$ (C) $\frac{3V}{K+2}$ (D) $\frac{3V}{K}$

Open solution

Q62. MCQ (Single Correct Answer)

A ball is thrown up with a certain velocity so that it reaches a height 'h'. Find the ratio of the two different times of the ball reaching $\frac{h}{3}$ in both the directions.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (B) $\frac{1}{3}$

(C) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ (D) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Open solution

Q63. MCQ (Single Correct Answer)

A 0.07 H inductor and a 12Ω resistor are connected in series to a 220V, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take π as $\frac{22}{7}$]

(A) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$ (B) 88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(C) 0.88 A and $\tan^{-1}\left(\frac{11}{6}\right)$ (D) 8.8 A and $\tan^{-1}\left(\frac{6}{11}\right)$

Open solution

Q64. MCQ (Single Correct Answer)

Two identical tennis balls each having mass 'm' and charge 'q' are suspended from a fixed point by threads of length 'l'. What is the equilibrium separation when each thread makes a small angle 'θ' with the vertical?

(A) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{2}}$

(B) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(C) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g}\right)^{\frac{1}{3}}$

(D) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g^2}\right)^{\frac{1}{3}}$

Open solution

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Q65. MCQ (Single Correct Answer)

Open solution

Assertion A : If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason R :

Least Count = $\frac{Pitch}{Total\ divisions\ on\ circular\ scale}$

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A.

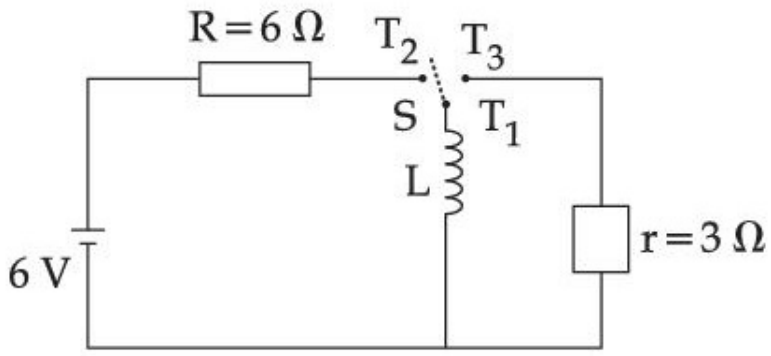
(C) A is correct but R is not correct.

(D) Both A and R are correct and R is NOT the correct explanation of A.

Q66. Numerical

Open solution

Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2. As the current in R = 6Ω attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across r = 3Ω resistor immediately after T_1 is connected to T_3 is _____ V. (Round off to the Nearest Integer)



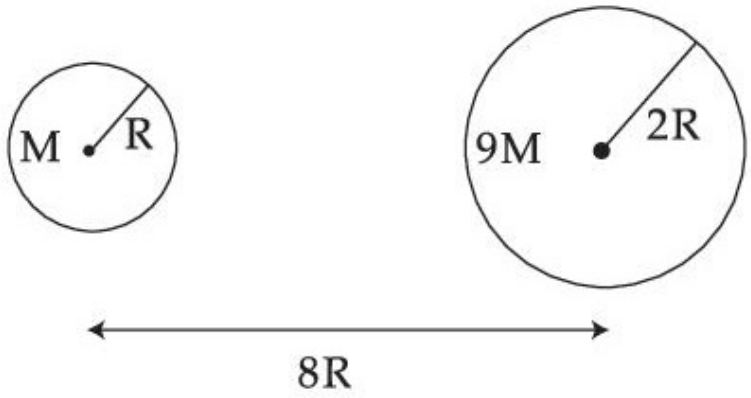
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Q67. Numerical

Open solution

Suppose two planets (spherical in shape) in radii R and 2R, but mass M and 9M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is _____.

[Given : The two planets are fixed in their position]



Q68. Numerical

Open solution

In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius 0.5 $\overset{o}{A}$. If the speed of electron is 2.2 $\times$ 10^6 m/s, then the current associated with the electron will be _____ $\times$ 10^{-2} mA. [Take π as $\frac{22}{7}$]

Q69. Numerical

Open solution

A particle of mass 9.1 $\times$ 10^{-31} kg travels in a medium with a speed of 10^6 m/s and a photon of a radiation of linear momentum 10^{-27} kg m/s travels in vacuum. The wavelength of photon is _____ times the wavelength of the particle.

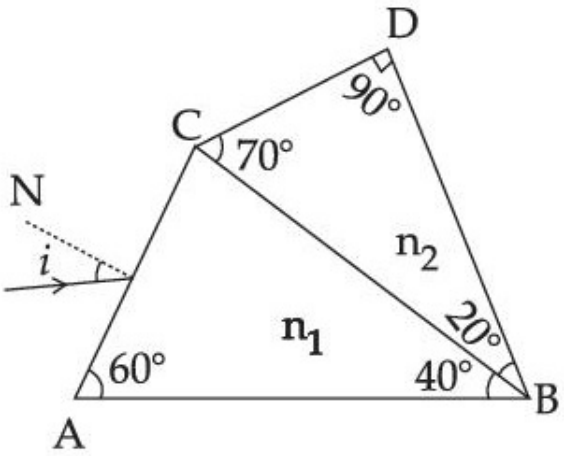
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Q70. Numerical

A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation

$n_1 = 1.2 + \frac{10.8\times10^{-14}}{\lambda^2}$ and $n_2 = 1.45 + \frac{1.8\times10^{-14}}{\lambda^2}$

The wavelength for which rays incident at any angle on the interface BC pass through without bending at that interface will be _____ nm.



Open solution

Q71. Numerical

A stone of mass 20 g is projected from a rubber catapult of length 0.1 m and area of cross section 10^{-6} m^2 stretched by an amount 0.04 m. The velocity of the projected stone is _____ m/s.

(Young's modulus of rubber = 0.5×10^9 N/m^2)

Open solution

Q72. Numerical

In a uniform magnetic field, the magnetic needle has a magnetic moment 9.85×10^{-2} A m^2 and moment of inertia 5×10^{-6} kg m^2. If it performs 10 complete oscillations in 5 seconds then the magnitude of the magnetic field is _____ mT. [Take π^2 as 9.85]

Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

The product obtained from the electrolytic oxidation of acidified sulphate solutions, is :

(A) HSO_4^- (B) HO_3SOOSO_3H (C) HO_2SOSO_2H (D) HO_3SOSO_3H

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

The parameters of the unit cell of a substance are a = 2.5, b = 3.0, c = 4.0, $\alpha = 90^\circ$, $\beta = 120^\circ$, $\gamma = 90^\circ$. The crystal system of the substance is :

(A) Hexagonal (B) Orthorhombic (C) Monoclinic (D) Triclinic

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following statements is NOT correct?

(A) Eutrophication indicates that water body is polluted

(B) The dissolved oxygen concentration below 6 ppm inhibits fish growth

(C) Eutrophication leads to increase in the oxygen level in water

(D) Eutrophication leads to anaerobic conditions

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.

Statement II : Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but statement II is true.

(B) Statement I is true but statement II is false.

(C) Both statement I and statement II are false.

(D) Both statement I and statement II are true.

Open solution

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II :

	List-I (Drug)		List-II (Class of Drug)
(a)	Furacin	(i)	Antibiotic
(b)	Arsphenamine	(ii)	Tranquilizers
(c)	Dimetone	(iii)	Antiseptic
(d)	Valium	(iv)	Synthetic antihistamines

Choose the most appropriate match :

(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The statement is incorrect about Ellingham diagram is

(A) provides idea about the reaction rate.

(B) provides idea about free energy change.

(C) provides idea about changes in the phases during the reaction.

(D) provides idea about reduction of metal oxide.

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : One is labelled as Assertion A and the other labelled as Reason R.

Assertion A : Lithium halides are some what covalent in nature.

Reason R : Lithium posses high polarisation capability.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and R are true but R is NOT the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

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Q80. Numerical · **Chemistry**

CO_2 gas adsorbs on charcoal following Freundlich adsorption isotherm. For a given amount of charcoal, the mass of CO_2 adsorbed becomes 64 times when the pressure of CO_2 is doubled. The value of n in the Freundlich isotherm equation is _____ $\times 10^{-2}$. (Round off to the Nearest Integer)

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

The value of $\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{j=1}^{n} \frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to :

(A) $5 + \log_e\left(\frac{3}{2}\right)$ (B) $2 - \log_e\left(\frac{2}{3}\right)$

(C) $3 + 2\log_e\left(\frac{2}{3}\right)$ (D) $1 + 2\log_e\left(\frac{3}{2}\right)$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The compound statement $(P \vee Q) \wedge (\sim P) \Rightarrow Q$ is equivalent to :

(A) $P \vee Q$ (B) $P \wedge \sim Q$

(C) $\sim (P \Rightarrow Q)$ (D) $\sim (P \Rightarrow Q) \Leftrightarrow P \wedge \sim Q$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let the plane passing through the point $(-1, 0, -2)$ and perpendicular to each of the planes $2x + y - z = 2$ and $x - y - z = 3$ be $ax + by + cz + 8 = 0$. Then the value of $a + b + c$ is equal to :

(A) 3 (B) 8 (C) 5 (D) 4

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Two tangents are drawn from the point P$(-1, 1)$ to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$. If these tangents touch the circle at points A and B, and if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is equal to :

(A) 2 (B) $(3\sqrt{2} + 2)$ (C) 4 (D) $3(\sqrt{2} - 1)$

Open solution

Q85. Numerical · **Mathematics**

Let a plane P pass through the point $(3, 7, -7)$ and contain the line, $\frac{x-2}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$. If distance of the plane P from the origin is d, then d^2 is equal to _____.

Open solution

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Q86. MCQ (Single Correct Answer) · **Physics**

If 'f' denotes the ratio of the number of nuclei decayed (N_d) to the number of nuclei at t = 0 (N_0) then for a collection of radioactive nuclei, the rate of change of 'f' with respect to time is given as :

[λ is the radioactive decay constant]

(A) $-\lambda(1-e^{-\lambda t})$

(B) $\lambda(1-e^{-\lambda t})$

(C) $\lambda e^{-\lambda t}$

(D) $-\lambda e^{-\lambda t}$

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

A body takes 4 min. to cool from 61° C to 59° C. If the temperature of the surroundings is 30° C, the time taken by the body to cool from 51° C to 49° C is :

(A) 4 min.

(B) 3 min.

(C) 8 min.

(D) 6 min.

Open solution

Q88. Numerical · **Physics**

A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 μF is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be _____ Ω.

Open solution

Q89. Numerical · **Physics**

A transistor is connected in common emitter circuit configuration, the collector supply voltage is 10 V and the voltage drop across a resistor of 1000 Ω in the collector circuit is 0.6 V. If the current gain factor (β) is 24, then the base current is _____ μA. (Round off to the Nearest Integer)

Open solution

Q90. Numerical · **Physics**

The amplitude of upper and lower side bands of A.M. wave where a carrier signal with frequency 11.21 MHz, peak voltage 15 V is amplitude modulated by a 7.7 kHz sine wave of 5V amplitude are $\frac{a}{10}V$ and $\frac{b}{10}V$ respectively. Then the value of $\frac{a}{b}$ is _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **C**	4. **B**	5. **D**
6. **C**	7. **D**	8. **C**	9. **D**	10. **C**
11. **B**	12. **B**	13. **A**	14. **5**	15. **12**
16. **15**	17. **19**	18. **2**	19. **108**	20. **31**
21. **1400**	22. **0**			

Mathematics

23. **D**	24. **B**	25. **B**	26. **A**	27. **B**
28. **C**	29. **C**	30. **C**	31. **D**	32. **C**
33. **A**	34. **D**	35. **D**	36. **C**	37. **C**
38. **C**	39. **5**	40. **2**	41. **3**	42. **1**
43. **6**	44. **16**	45. **490**	46. **2**	47. **5**

Physics

48. **A**	49. **C**	50. **C**	51. **C**	52. **D**
53. **C**	54. **A**	55. **D**	56. **B**	57. **B**
58. **A**	59. **C**	60. **D**	61. **C**	62. **C**
63. **A**	64. **B**	65. **A**	66. **3**	67. **4**
68. **112**	69. **910**	70. **600**	71. **20**	72. **8**

Out of syllabus

73. **B**	74. **C**	75. **C**	76. **D**	77. **D**
78. **A**	79. **D**	80. **17**	81. **D**	82. **D**
83. **D**	84. **C**	85. **3**	86. **C**	87. **D**
88. **150**	89. **25**	90. **1**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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